

Early Behavioral Detection of Cryptojacking Malware Using Multi-Layer System Telemetry and Machine Learning*

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Abstract—Cryptojacking is now a serious cybersecurity issue where attackers misuse compromised computing resources to perform cryptocurrency mining without the system owner being aware of it. Several of the existing detection solutions primarily focus on identifying unusual CPU consumption or irregular network communication only after the mining process has already started, which is often too late. As a result of this, such methods often struggle to recognize cryptojacking activities during the early stages of infection, allowing malicious operations to run tenaciously for longer durations.

This research introduces an early behavioral detection framework intended to identify cryptojacking malware by leveraging multi-layer system telemetry together with machine learning techniques. The framework evaluates behavioral indicators collected from multiple layers of system activity, including process execution behavior, CPU utilization characteristics, and patterns of network communication. By integrating and correlating these telemetry signals, the system builds behavioral profiles that help distinguish legitimate system operations from activities associated with concealed cryptocurrency mining.

Before long-term mining operations are established entirely, machine learning-based anomaly detection models are used to identify deviations from regularly expected behavioral patterns and signal possible cryptojacking activity. In comparison to traditional single-layer monitoring methods, experimental analysis using cybersecurity datasets shows that deploying a multi-layer behavioral strategy simultaneously reduces false positive rates and enhances early detection performance. These results highlight the usefulness of integrating machine learning methods with system telemetry analysis for proactive detection of cryptojacking threats.

Index Terms—Cryptojacking, Malware Detection, Cybersecurity, Behavioral Analysis, Machine Learning, System Telemetry, Anomaly Detection

I. INTRODUCTION

The rapid growth of cryptocurrency technologies has opened new possibilities for decentralized financial transactions while

also introducing an array of emerging cybersecurity challenges in modern day to day computing environments. Among these threats, cryptojacking has gained increased attention. Cryptojacking refers to a cyber attack in which attackers secretly utilize the computational resources of devices by intrusion to perform cryptocurrency mining without the awareness or consent of the system owner, and often goes undetected. In comparison to traditional strains of malware that are steal sensitive information or disrupt system operations, cryptojacking malware typically runs quietly in the background while continuously consuming various system resources such as processing power and electrical energy. As the popularity and market value of cryptocurrencies are seeing a continuous rise, attackers have gained more motivation to exploit vulnerable systems for the purpose of unauthorized mining [1][2].

Numerous computing environments such as individual workstations, corporate infrastructures, cloud-based platforms, and Internet-of-things (IoT) devices are highly vulnerable to cryptojacking attacks. Attackers frequently use malicious scripts embedded in compromised websites, infected software distributions, or malware programs that install hidden mining applications on targeted systems to carry out cryptojacking. Attackers have altered trustworthy mining software, including programs like XMRig to operate surreptitiously within compromised systems, according to a number of documented incidents. It can be a tedious and difficult task to differentiate these mining processes from regular system operations using traditional security techniques because they appear to be legitimate computational workloads [7][8].

Traditional methods for detecting cryptojacking depend on monitoring system indicators like abnormally high CPU utilization, erratic power consumption patterns, or suspicious communication with external cryptocurrency mining pools.

The drawback of these techniques is that they usually only identify the attack after the mining process has been operating for a significant amount of time. Although these indicators might show the existence of active mining activity, by then the target device might already be experiencing reduced operational efficiency, excessive energy consumption, and deteriorated performance. This restriction highlights the imminent need to create proactive detection systems that can spot cryptojacking activity in its early stages before significant system damage occurs [3][8].

Recent advances in machine learning have seen enhanced malware detection and strengthened cybersecurity monitoring frameworks. Machine learning algorithms are capable of discovering patterns that distinguish normal operational behavior from suspected malicious activity. In the context of cryptojacking detection, potential indicators tend to appear across several layers of system activity including process creation patterns, CPU usage characteristics and network communication behavior. When these signals are examined, machine learning techniques have the ability to reveal subtle anomalies associated with cryptojacking activities that may go undetected through conventional monitoring approaches [9][15].

Despite these advancements, most of the ongoing studies concentrate on single-layer monitoring techniques, like independent examination of CPU utilization or network traffic patterns. Although these methods are capable of identifying specific mining activities, they might not be able to identify the connections between various system behaviors. Because of this restriction, they are less effective against sophisticated or covert cryptojacking attacks that aim to avoid detection by conventional means. To gain a more comprehensive understanding of system behavior, a more thorough approach integrates telemetry data from various levels of system activity. By identifying intricate connections between processes, resource usage trends, and network interactions, multi-layer behavioral analysis enhances the capacity to identify cryptojacking attempts early on. [3][9].

In this work, we present an early behavioral detection framework that combines machine learning methods with multi-layer system telemetry to detect cryptojacking malware. The suggested framework collects system telemetry data from multiple sources such as network communication characteristics, CPU utilization statistics and process-level activity. After processing these telemetry signals, behavioral characteristics that depict patterns of system activity over time are extracted. After that, machine learning-based anomaly detection models are used to distinguish between suspicious activity that might point to the existence of hidden cryptojacking malware and typical operational behavior.

The main aim of conducting this research is to ensure that it is possible to detect cryptojacking activities in their initial stages, even before the mining process is fully integrated into the compromised system. This will ensure that resources are not exploited for a longer time and the impact of such attacks is reduced. In addition, the use of machine learning algorithms will ensure that the detection system improves with time as it

adapts to changing attack patterns.

The main contributions of this paper can be listed as follows. Firstly, a multi-layer telemetry monitoring framework for observing system behaviors associated with cryptojacking activities is proposed. Secondly, a machine learning-based detection model is developed for analyzing the behavioral patterns at different layers of a system in order to identify the activities associated with cryptojacking and detect it at an early stage. Finally, the performance of the proposed framework is validated through experimental evaluation ensuring desired performance.

The rest of this paper is organized as follows. In Section II, the relevant literature on cryptojacking detection and machine learning-based malware analysis is discussed. In Section III, the proposed detection method and overall system architecture are presented. In Section IV, the proposed system architecture along with the telemetry collection mechanisms is discussed. In Section V, the dataset preparation along with the data preprocessing methods used in the proposed work is discussed. In Section VI, the experiment setup along with the evaluation methods used in the proposed work is discussed. In Section VII, the experiment results along with the performance analysis are presented. In Section VIII, the overall paper is concluded along with the possible future work.

II. RELATED WORK

The phenomenon of cryptojacking has gained significant research attention in recent times due to the rise in the usage of computer resources for carrying out illicit practices of cryptomining. Several research works have been done on the growing trend of cryptojacking attacks performed either through browsers or malware. The potential impact of these attacks on the current computing environment has been a subject of interest for many researchers [1][7]. Several approaches to detect these attacks have been put forward. The existing detection mechanisms are mostly based on the monitoring of system resources and the use of machine learning.

The initial research focus in the detection of cryptojacking activities was mainly centered on the identification of unusual patterns in the utilization of system resources. Due to the high computation requirements for cryptocurrency mining activities, some detection mechanisms focus on identifying unusually high CPU utilization patterns or unusual patterns in power consumption activities [8]. The mechanisms have the capability to identify mining activities that are already in progress. However, the mechanisms face difficulties in detecting the activities at earlier stages. At these stages, the activities can still be similar to normal system activities.

Another form of analysis that is commonly carried out is the analysis of the network traffic that is involved in the communication with the cryptocurrency mining pools. The malware that is involved in the mining of the cryptocurrency usually communicates with the external servers for the reception of the mining tasks as well as the sending of the results that are generated from the completion of the computational tasks. The monitoring systems that are based on the network usually seek

to detect the suspicious patterns of the communication that is involved in the mining of the cryptocurrency [3]. However, the attackers usually employ various countermeasures that make the detection mechanisms based on the analysis of the network traffic less effective.

More recent studies have also investigated the possibility of applying machine learning for malware detection and cryptojacking attacks. Machine learning algorithms are capable of processing vast amounts of data generated by computer systems through various telemetry data sources, allowing them to identify behavioral patterns that differentiate normal system operations from possible malicious activities [9]. Various machine learning algorithms, including supervised and unsupervised algorithms, have been utilized for detecting abnormal resource consumption patterns and behavioral characteristics of cryptojacking attacks [15].

Progress has been made, yet most existing detection methods still rely on monitoring a single layer of system behavior such as CPU utilization or network communication. These approaches fall short when it comes to capturing complex behavioral relationships exhibited by modern cryptojacking malware, which has high stealth capabilities and can evade traditional security mechanisms [3]. Integration of telemetry information collected from multiple system layers therefore provides a far more comprehensive representation of system activity and improves the accuracy of early cryptojacking detection.

In order to overcome these limitations, our study proposes a multi-layer behavioral detection approach which considers the integration of telemetry information from process activity, CPU usage patterns, and network communication features. By combined analysis of the above mentioned features, the current approach can identify cryptojacking behavior at the early stages with a much lesser rate of false positive alarms.

TABLE I: Comparison of Existing Cryptojacking Detection Approaches

Detection Method	Data Source	Technique	Limitation
CPU Usage Monitoring	System Logs	Resource Monitoring	Detects attacks only after mining begins
Network Traffic Analysis	Network Packets	Traffic Inspection	Encrypted traffic reduces detection accuracy
Machine Learning	Resource Usage Data	Pattern Classification	Limited feature diversity
Multi-layer Behavioral Analysis	System Telemetry	Behavioral Modeling	Early-stage detection capability

III. PROPOSED METHODOLOGY

This section describes the proposed framework designed to detect cryptojacking malware at an early stage by leveraging multi-layer system telemetry together with machine learning techniques. The framework observes behavioral signals generated from different layers of system activity and evaluates them

collectively to identify patterns that may suggest unauthorized cryptocurrency mining. By integrating telemetry information obtained from multiple system sources, the framework aims to detect suspicious behavior before prolonged mining operations begin to significantly affect system performance.

The overall detection framework consists of four primary stages: telemetry data collection, feature extraction, multi-layer behavioral analysis, and machine learning-based anomaly detection. Combining telemetry signals from several layers of system activity allows the framework to obtain a broader and more detailed representation of system behavior, which improves the reliability and effectiveness of cryptojacking detection [3][9].

A. System Telemetry Collection

The first stage of the proposed framework involves collecting system telemetry data that reflects the operational behavior of the monitored computing environment. Cryptojacking malware often demonstrates distinctive behavioral patterns during execution, particularly in terms of resource consumption characteristics and persistent background process activity.

In this study, telemetry information is gathered from multiple system components, including CPU utilization statistics, system process activity, and network communication characteristics. CPU usage metrics provide useful insight into abnormal computational workloads that may indicate the existence of hidden mining processes. Process-level telemetry captures events such as process creation, execution duration, and parent-child process relationships, which can reveal suspicious background activity. In addition, network telemetry records communication between the host system and external servers, including connections that may correspond to cryptocurrency mining pools or command-and-control infrastructure used by attackers [8].

Collecting telemetry signals across several layers of system activity enables the framework to develop a more comprehensive understanding of system behavior and improves its ability to identify indicators associated with cryptojacking activity.

B. Feature Extraction

Once telemetry data has been collected, relevant behavioral attributes are extracted and organized into a structured format suitable for machine learning analysis. Feature extraction converts raw telemetry measurements into quantifiable indicators that represent both normal operational patterns and potential anomalous activity. Examples of extracted features include average CPU utilization, fluctuations in CPU usage over time, the number of active processes, the execution duration of processes, and the frequency of outbound network communications. These features provide valuable insights into the system's computational workload and process behavior while helping identify deviations from normal operational patterns.

To ensure consistency during the training phase, feature normalization techniques are applied to scale different telemetry attributes into comparable numerical ranges. This preprocessing step improves the stability, convergence speed, and

predictive performance of the machine learning models used within the detection framework [15].

C. Multi-Layer Behavioral Analysis

In contrast to traditional detection approaches that rely on a single indicator such as CPU usage or network activity alone, the proposed framework performs behavioral analysis by examining multiple telemetry layers simultaneously. This multi-layer analysis allows the system to identify correlations between various behavioral indicators. For example, a sudden increase in CPU utilization combined with the creation of persistent background processes and frequent communication with external servers may strongly suggest the presence of cryptojacking activity. Evaluating these signals collectively enables the framework to detect complex behavioral patterns that may remain undetected when monitoring individual telemetry indicators independently.

This behavioral modeling approach increases the reliability of the detection mechanism and reduces the probability of false positives that might otherwise be caused by legitimate high-performance workloads or temporary spikes in system resource usage.

D. Machine Learning Detection Model

The final stage of the proposed framework applies machine learning techniques to classify system behavior and identify anomalies that may indicate cryptojacking attacks. Machine learning models are trained using the extracted telemetry features in order to distinguish between normal operational behavior and suspicious activity patterns.

In this study, anomaly detection techniques are utilized to identify deviations from baseline system behavior. These models learn the normal characteristics of system activity and subsequently flag unusual behavioral patterns that may correspond to unauthorized cryptocurrency mining operations.

By integrating machine learning methods with multi-layer telemetry analysis, the proposed detection framework is capable of adapting to evolving attack strategies and identifying cryptojacking activity during its early stages. This capability enhances detection accuracy while maintaining scalability and practical applicability for deployment in real-world computing environments [9][15].

IV. SYSTEM ARCHITECTURE

This section describes the architecture of the proposed cryptojacking detection system. The architecture is designed to identify unauthorized cryptocurrency mining activities by analyzing behavioral signals obtained from multiple layers of system telemetry. Unlike conventional security mechanisms that rely on a single monitoring indicator, the proposed architecture integrates several monitoring components into a unified detection pipeline. The framework combines system monitoring, feature extraction, behavioral modeling, and machine learning-based detection to continuously analyze system activity and identify suspicious patterns associated with cryptojacking attacks.

By collecting telemetry signals from different system layers, the proposed architecture captures a comprehensive representation of system behavior. This multi-layer monitoring strategy enables the system to identify abnormal resource utilization patterns that may indicate the presence of hidden cryptocurrency mining processes. Integrating telemetry from process activity, CPU utilization, and network communication provides richer contextual information about system behavior, thereby improving the reliability and accuracy of detection mechanisms [3][9].

As illustrated in Fig. 1, the proposed architecture consists of four primary components: the telemetry monitoring layer, feature extraction and representation module, multi-layer behavioral analysis module, and the machine learning detection engine. These components operate sequentially and form a continuous monitoring pipeline that analyzes system activity in real time. Each stage of the architecture contributes to transforming raw telemetry signals into meaningful behavioral insights that can be used to detect anomalous activities associated with cryptojacking malware.

A. Telemetry Monitoring Layer

The telemetry monitoring layer forms the foundation of the proposed detection architecture. Its primary objective is to continuously collect system activity data from the monitored computing environment. Cryptojacking malware typically generates distinctive behavioral patterns during execution, particularly in relation to abnormal CPU consumption, persistent background processes, and communication with external mining servers.

To capture these behavioral indicators, the telemetry monitoring layer gathers information from multiple system components. CPU utilization statistics are monitored to identify unusual computational workloads that may indicate hidden mining processes running in the background. In addition to CPU usage, the monitoring system collects process-level telemetry data including process creation events, execution duration, process hierarchy relationships, and resource consumption metrics.

Monitoring process activity is particularly important because many cryptojacking malware variants attempt to disguise mining processes as legitimate system tasks. Observing process-level attributes such as long-running background processes or abnormal process spawning patterns can therefore reveal suspicious activities that might otherwise remain unnoticed.

The telemetry monitoring layer also captures network communication information. Cryptojacking malware frequently communicates with remote cryptocurrency mining pools or command-and-control servers to obtain mining tasks and submit computed hashes. Therefore, monitoring outbound network connections, packet characteristics, and communication frequency can provide valuable insights into potential mining behavior [8].

By continuously collecting telemetry information from these multiple sources, the monitoring layer provides a comprehen-

sive dataset describing the operational state of the system. This dataset forms the input for subsequent stages of the detection pipeline.

B. Feature Extraction and Representation

After telemetry signals are collected, the raw system data is processed through the feature extraction module. The purpose of this stage is to convert raw telemetry measurements into structured behavioral attributes that can be used as inputs for machine learning algorithms.

Raw telemetry data often contains noise, redundant measurements, and heterogeneous attributes that are not directly suitable for model training. Feature extraction therefore focuses on transforming this data into meaningful indicators that capture important characteristics of system behavior. Typical features extracted in this stage include average CPU utilization over time, variance in CPU usage patterns, number of active processes, duration of long-running processes, and frequency of process creation events.

Network-related behavioral attributes are also extracted, including the number of outbound network connections, packet length distributions, and flow-level communication characteristics. These attributes are important because cryptojacking malware commonly communicates with mining pools to exchange computational tasks and transmit mining results [3].

Once these behavioral attributes are extracted, normalization and standardization techniques are applied to ensure that different features are scaled into comparable numerical ranges. Feature normalization improves model stability, reduces training bias caused by feature magnitude differences, and enhances the overall predictive performance of machine learning algorithms used in the detection framework [15].

The resulting feature vectors represent a structured behavioral profile of system activity and are forwarded to the behavioral analysis module for further evaluation.

C. Multi-Layer Behavioral Analysis

The multi-layer behavioral analysis module examines relationships between different telemetry features in order to identify suspicious behavioral patterns. Traditional cryptojacking detection techniques often rely on monitoring a single indicator such as high CPU usage or abnormal network traffic. However, such isolated indicators may not always reliably indicate malicious activity because legitimate high-performance applications can produce similar system behavior.

To overcome this limitation, the proposed architecture analyzes multiple telemetry layers simultaneously. By correlating CPU utilization patterns with process behavior and network communication characteristics, the system can detect subtle relationships that are indicative of cryptojacking activity.

For example, a sudden increase in CPU utilization combined with persistent background processes and repeated communication with external mining servers may strongly indicate unauthorized mining activity. Individually, these signals may not necessarily appear suspicious. However, when evaluated

together, they form a behavioral pattern that suggests the presence of hidden mining malware.

This multi-layer correlation strategy enables the system to detect complex behavioral signatures associated with modern cryptojacking attacks. Additionally, evaluating multiple indicators simultaneously reduces the likelihood of false positive detections caused by legitimate resource-intensive workloads.

D. Machine Learning Detection Engine

The final component of the proposed architecture is the machine learning detection engine. This module analyzes the extracted behavioral features and determines whether the observed system activity deviates from normal operational patterns.

Machine learning models are particularly well suited for analyzing complex telemetry data because they can learn hidden relationships within large datasets. In the proposed framework, multiple machine learning algorithms are employed to evaluate system behavior, including Random Forest, XGBoost, LightGBM, CatBoost, and Support Vector Machine classifiers.

During the training phase, these models learn baseline patterns representing normal system activity. Once trained, the models can analyze new telemetry data and identify deviations that may correspond to cryptojacking behavior. When anomalous patterns are detected, the system flags the activity as potentially malicious and generates an alert for further investigation.

The integration of machine learning techniques with multi-layer telemetry analysis significantly enhances the detection capability of the proposed system. By combining behavioral monitoring with adaptive learning models, the framework is able to detect cryptojacking activity at earlier stages while maintaining scalability and efficiency. This approach makes the system suitable for deployment in modern computing environments where large volumes of system telemetry data must be analyzed continuously [9][15].

E. Alerts and Feedback Mechanism

In addition to detecting suspicious activity, the proposed system incorporates an alert and feedback component that notifies administrators when potential cryptojacking behavior is identified. When the detection engine flags an anomaly, the system generates alerts containing relevant information such as the affected process, abnormal resource usage patterns, and associated network connections. These alerts assist security analysts in quickly investigating and responding to possible threats. The feedback mechanism also enables administrators to review detection results and provide confirmation regarding whether an alert corresponds to malicious activity or a benign system event. This feedback can be used to refine detection thresholds and improve the accuracy of the monitoring system over time.

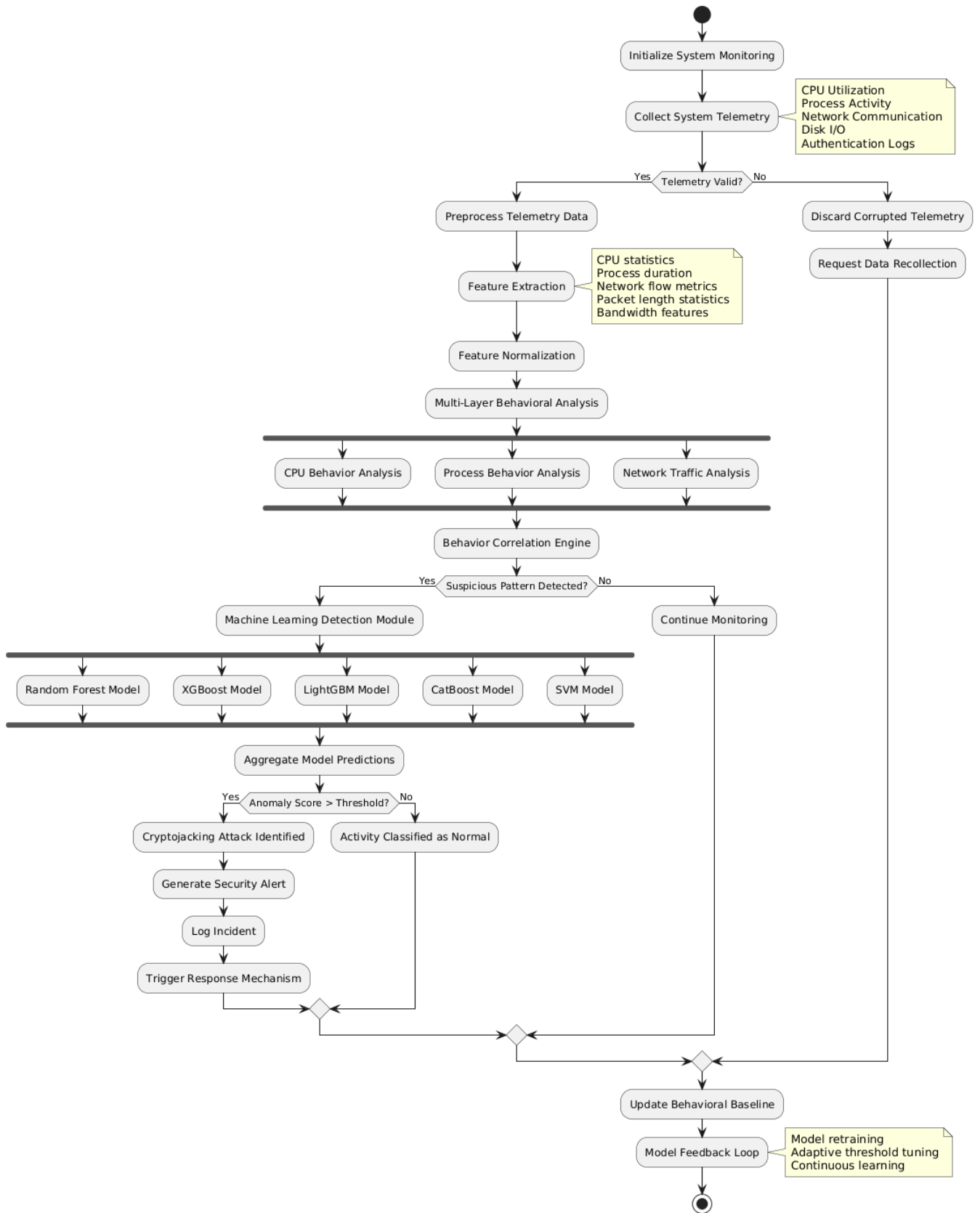


Fig. 1: Proposed multi-layer behavioral detection framework for early identification of cryptojacking malware. The architecture integrates telemetry collection, feature extraction, multi-layer behavioral analysis, and machine learning-based anomaly detection to identify unauthorized cryptocurrency mining activities.

V. DATASET AND PREPROCESSING

The performance of the proposed cryptojacking detection framework relies on the availability of accurate and consistent system telemetry data. The dataset utilized in this study is composed of system activity logs that record several behavioral indicators related to computational resource utilization and process execution within the operating environment.

A. Dataset Description

The dataset includes telemetry information collected from monitored computing systems. These telemetry signals consist of CPU utilization statistics, process execution details, memory usage patterns, and records of network communication activity. Such information provides a detailed view of how system resources are consumed over time and supports the identification of unusual behavior that may suggest unauthorized cryptocurrency mining operations. Each entry in the dataset represents a snapshot of system activity observed during a defined time interval. The collected telemetry data is organized into multiple attributes that describe different operational aspects of the system, including resource usage patterns, characteristics of running processes, and interactions with network services.

B. Data Preprocessing

Before the detection model is trained, the collected telemetry information is processed through several preprocessing steps to improve data quality and ensure consistency. These preprocessing procedures include cleaning the dataset, normalizing feature values, and organizing the telemetry records into structured time intervals. Data cleaning focuses on identifying and removing incomplete, inconsistent, or corrupted records that could negatively influence the performance of the learning model. Normalization methods are then applied to scale numerical attributes so that variables with larger value ranges do not disproportionately influence the training process.

C. Feature Engineering

Feature engineering is applied to convert raw telemetry information into structured behavioral indicators that describe the operational condition of the system. Several representative attributes are extracted from the dataset, including average CPU utilization, variation in CPU consumption over time, the number of active processes, execution duration of long-running tasks, and the frequency of outbound network connections. By examining these indicators, the proposed framework can distinguish normal system operations from potentially suspicious activity linked to cryptojacking attacks.

VI. EXPERIMENTAL SETUP

This section outlines the experimental configuration used to evaluate the proposed cryptojacking detection framework. The primary objective of the experiment is to analyze telemetry data generated by computing systems and identify behavioral patterns that may indicate unauthorized cryptocurrency mining activity.

A. Implementation Environment

The proposed detection framework is implemented within a data analysis and machine learning environment capable of processing large volumes of telemetry information. The system processes activity logs collected from monitored computing systems and extracts behavioral features that describe system operation.

The implementation pipeline consists of several modules responsible for telemetry processing, feature extraction, and anomaly detection. Machine learning algorithms are applied to the extracted features to identify deviations from typical system behavior.

B. Model Training

During the training stage, the machine learning models learn the baseline behavioral characteristics associated with normal system operation. The training dataset contains telemetry records that represent legitimate patterns of system usage. The models analyze behavioral attributes such as CPU utilization trends, number of active processes, process execution duration, and frequency of network communication events.

After learning these baseline patterns, the trained models can identify deviations that may correspond to suspicious activity related to cryptojacking.

C. Evaluation Metrics

To assess the effectiveness of the proposed detection framework, several widely used machine learning evaluation metrics are employed. These metrics measure how well the models differentiate between normal system behavior and potentially malicious activity.

The following metrics are considered for performance evaluation:

- **Accuracy** – Indicates the overall proportion of correctly classified observations.
- **Precision** – Represents the fraction of detected threats that are actually malicious.
- **Recall** – Measures how effectively the system identifies real cryptojacking instances.
- **F1-Score** – Provides a balanced evaluation by combining precision and recall into a single metric.

Together, these metrics provide a comprehensive evaluation of the performance of the proposed cryptojacking detection framework.

VII. RESULTS AND PERFORMANCE EVALUATION

This section will assess the proposed framework for cryptojacking detection and its performance. The aim is to assess how effective the framework is in detecting unusual behaviors that may indicate unauthorized cryptocurrency mining on a host machine. The framework analyzes the telemetry data collected from various operational layers, including CPU usage, process activity, and network information. The framework combines these signals to identify unusual system activity and point to possible cryptojacking events.

A. Detection Performance

The machine learning part of the system evaluates behavioral features derived from system telemetry data and assigns an anomaly score to each monitored activity pattern. Each score represents how much the observed activity differs from the baseline profile established during normal system operation.

The results indicate that the multi-layer telemetry method is more reliable than the single-layer method in cryptojacking detection. Since it considers multiple indicators of behavior, the system is able to identify small patterns that are likely to be overlooked when viewed individually.

B. Performance Metrics Analysis

The performance of the proposed detection system is measured using a set of accepted machine learning metrics, including accuracy, precision, recall, and F1-score. The four metrics collectively offer a complete insight into the detection capability of the proposed system. Accuracy is defined as the ratio of correctly classified instances to the total number of instances used for testing.

Precision is the ratio of correctly identified instances of cryptojacking to the total number of instances identified as potential threats. Recall is the ratio of correctly identified instances of cryptojacking to the total number of instances of cryptojacking. The F1-score is a balanced measure of precision and recall.

The experimental outcome indicates that the inclusion of telemetry data from multiple layers of behavioral information has a substantial impact on enhancing detection capability and minimizing the possibility of false positives.

C. Comparative Discussion

When compared with the conventional detection methods that depend only on the monitoring of resource utilization or network traffic analysis, the proposed framework shows better detection capability by combining the telemetry data collected from various levels of the system. The proposed combined monitoring approach enables the system to detect cryptojacking activity at an earlier stage and provides better protection against stealthy cryptocurrency mining malware.

From the experimental results, it is confirmed that the proposed machine learning-based detection framework is effective in detecting cryptojacking activity while maintaining a reliable monitoring process.

D. Model Performance Comparison

A comparison of various machine learning models for the cryptojacking detection task is presented in Table II. The machine learning models used for comparison include ensemble learning models like Random Forest, XGBoost, LightGBM, and CatBoost, and the traditional Support Vector Machine (SVM) classifier. The comparison of the models is done using various metrics such as accuracy, precision, recall, F1-score, Area Under the Curve (AUC), and average precision. These

metrics give a complete insight into the classification strength and reliability of the models used for comparison.

From the results shown in Table II, it is clear that ensemble learning models perform better than the traditional Support Vector Machine classifier for cryptojacking detection. This is because ensemble learning models use the predictions of multiple weak models to build a strong predictive model. This makes it easier for the models to identify the complex nonlinear patterns in system telemetry data, which is particularly useful in cryptojacking detection since malicious activity is usually represented by small behavioral patterns scattered across multiple system attributes.

Among the algorithms used, the Random Forest classifier had the highest accuracy of 0.99717. The Random Forest algorithm works by creating a number of decision trees and then using their predictions to come up with the final result. The algorithm is stable due to its bagging approach, which helps it to work well even when the data is noisy or of high dimensionality. The high precision and recall scores of the Random Forest algorithm show that it is able to accurately distinguish between cryptojacking attacks and normal system activity with a high degree of accuracy and with very few errors.

Boosting algorithms such as XGBoost, LightGBM, and CatBoost also performed well in terms of predictive accuracy. The XGBoost algorithm had an accuracy of 0.99699 and a precision of 1.00, which shows that the algorithm is very accurate in its detection of cryptojacking attacks and that it makes very few false positives. The gradient boosting approach of the XGBoost algorithm helps it to improve its decision trees by giving more weight to the samples that were previously misclassified.

The LightGBM model was also competitive in terms of performance for all evaluation metrics and had the highest AUC of 0.99986 among all the algorithms used. LightGBM uses a histogram-based gradient boosting algorithm that improves the computational efficiency of the model without compromising its predictive power. Its capability to handle large datasets with a moderate memory footprint makes it an ideal choice for a telemetry analysis and threat detection setup.

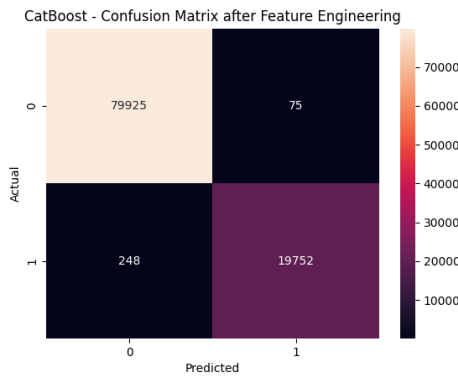
The CatBoost model was also competitive in terms of classification performance with high precision, recall, and F1-score values.

E. Model Evaluation Results

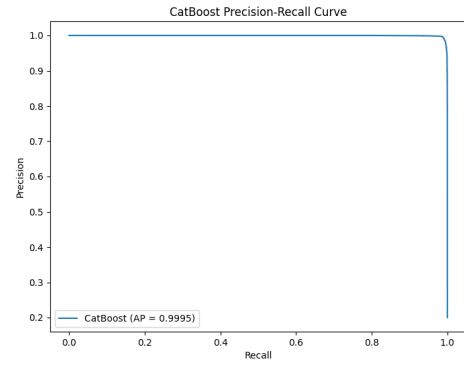
In order to further analyze the behavior of the trained machine learning models, some visual assessment metrics were created, such as confusion matrices, ROC curves, and learning curves. These provide additional information about the classification performance and generalization capability of the models.

TABLE II: Performance Comparison of Machine Learning Models for Cryptojacking Detection

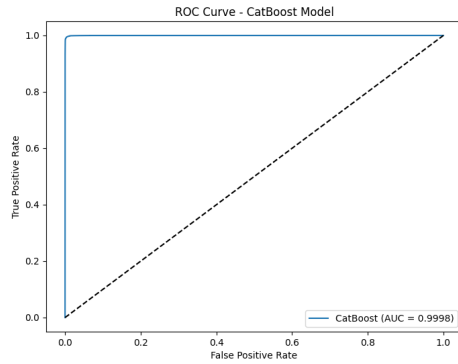
Model	Algorithm Type	Accuracy	Precision	Recall	F1 Score	AUC	Avg Precision	Key Characteristics
Random Forest	Bagging Ensemble of Decision Trees	0.99717	0.99	0.99	0.99	0.99916	0.99721	Handles nonlinear data well and robust to noise
XGBoost	Gradient Boosted Decision Trees	0.99699	1.00	0.99	0.99	0.99984	0.99961	High predictive performance with optimized boosting
LightGBM	Histogram-based Gradient Boosting	0.99694	1.00	0.99	0.99	0.99986	0.99961	Efficient training with reduced memory usage
CatBoost	Ordered Gradient Boosting	0.99677	1.00	0.99	0.99	0.99983	0.99949	Handles categorical features effectively
SVM	Linear Margin-based Classifier	0.89282	0.69	0.83	0.76	0.95899	0.87856	Linear decision boundary with lower complexity



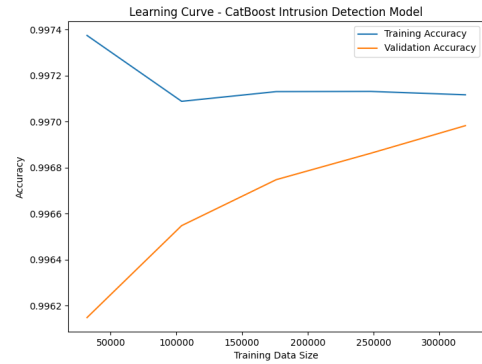
(a) Confusion Matrix



(a) Precision-Recall Curve



(b) ROC Curve

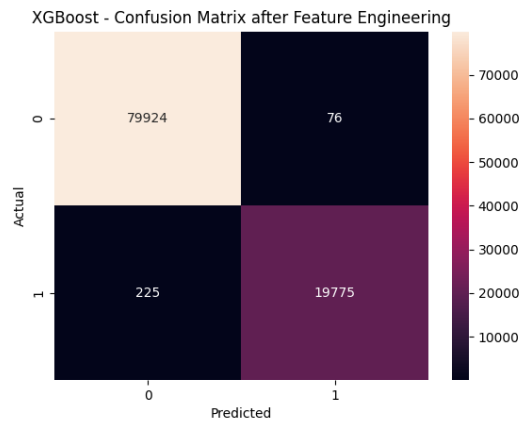


(b) Learning Curve

Fig. 2: Evaluation results of the CatBoost model (Part 1).

Fig. 3: Evaluation results of the CatBoost model (Part 2).

The performance of the XGBoost model was tested using various evaluation metrics, such as the confusion matrix, ROC curve, precision-recall curve, and learning curve. XGBoost is a type of gradient boosting algorithm that builds an additive model of a collection of additive decision trees. The above-mentioned graphs in Figures 4 and 5 illustrate the classification accuracy and generalization ability of the XGBoost model.



(a) Confusion Matrix

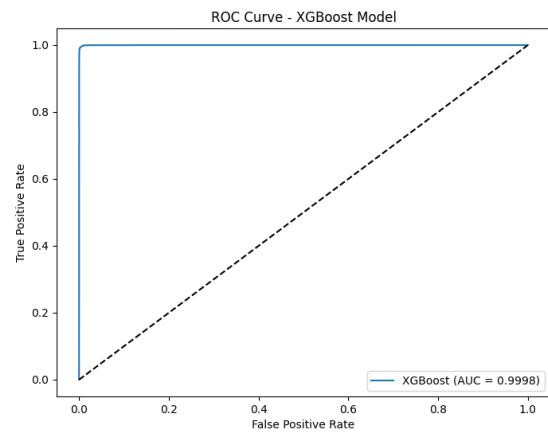
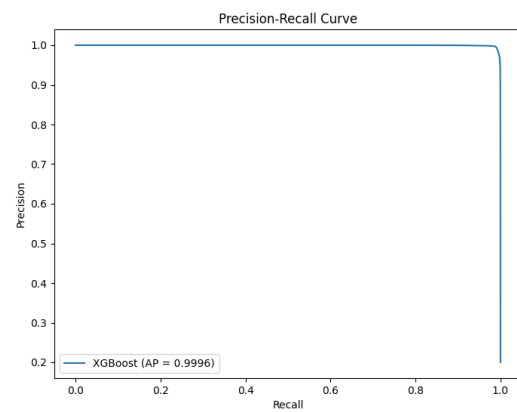
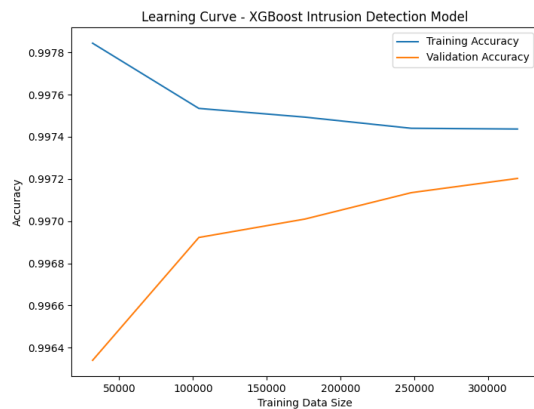


Fig. 5: ROC Curve for the XGBoost model (Part 2).



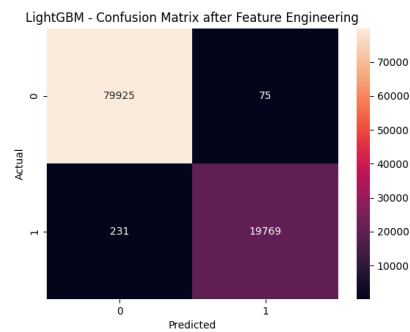
(b) Precision-Recall Curve



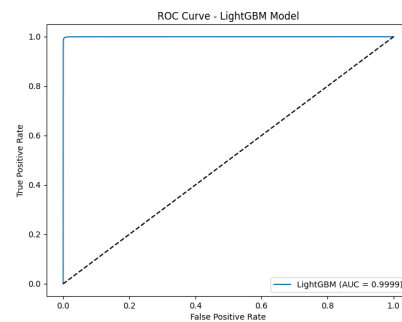
(c) Learning Curve

Fig. 4: Evaluation metrics for the XGBoost model (Part 1).

The LightGBM model was also analyzed using a number of visualization-based evaluation metrics such as the confusion matrix, ROC curve, precision-recall curve, and learning curve. LightGBM is a histogram-based gradient boosting framework that supports efficient training while maintaining high predictive accuracy. The graphs shown in Figures 6 and 7 illustrate the classification accuracy and generalization performance of the model.

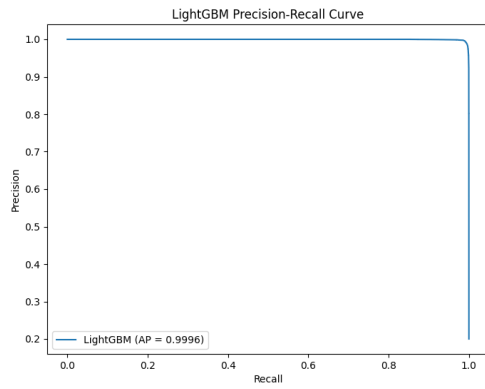


(a) Confusion Matrix

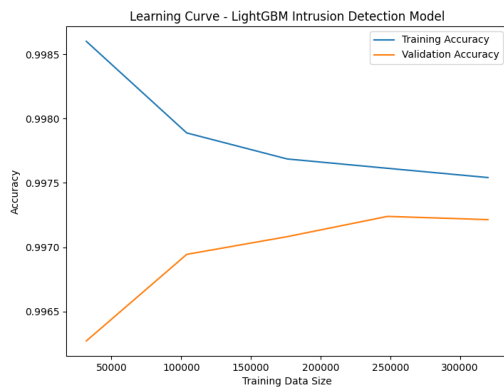


(b) ROC Curve

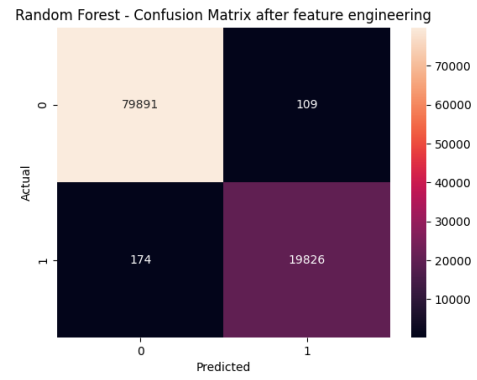
Fig. 6: Evaluation results of the LightGBM model (Part 1).



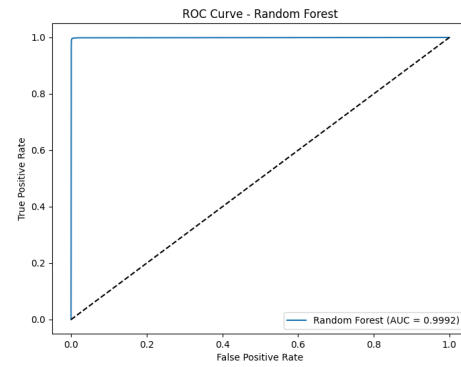
(a) Precision-Recall Curve



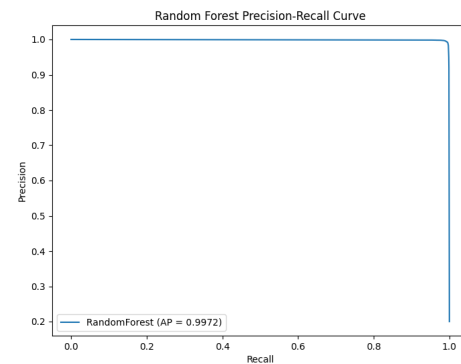
(b) Learning Curve



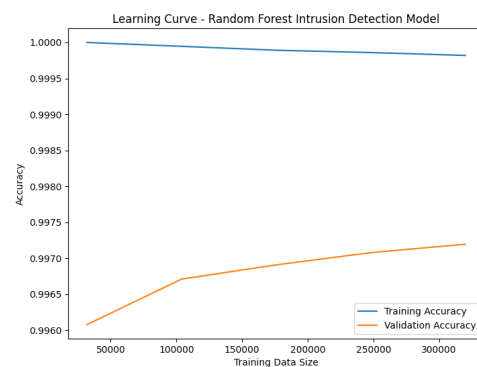
(a) Confusion Matrix



(b) ROC Curve



(c) Precision-Recall Curve



(d) Learning Curve

Fig. 7: Evaluation results of the LightGBM model (Part 2).

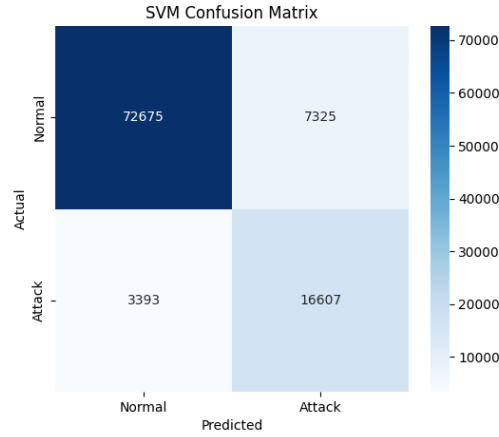
The Random Forest model was evaluated using various statistical evaluation metrics combined with visualization analysis tools such as the confusion matrix, ROC curve, precision-recall curve, and learning curve. Random Forest is an ensemble learning method that produces many decision trees during the training process and combines their predictions to improve the accuracy of classification. The ensemble learning method reduces the chances of overfitting the data and allows the model to learn complex nonlinear relationships that exist in the system telemetry feature space.

Based on the evaluation metrics summarized in Table II, the Random Forest classifier achieved the highest overall accuracy of 0.99717 with precision and recall of 0.99, thereby achieving an F1-score of 0.99. These results indicate that the model is accurate in identifying legitimate system activity and cryptojacking events. Additionally, the model's discriminative power is excellent, as indicated by the AUC value of 0.99916 in the ROC and precision-recall curves, thereby achieving an almost perfect separation between malicious and legitimate data points.

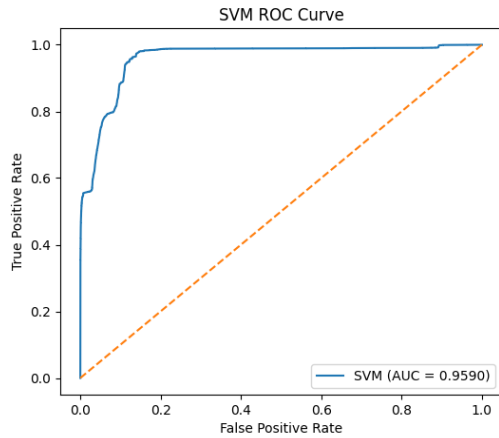
Fig. 8: Evaluation results of the Random Forest model.

The Support Vector Machine (SVM) classifier was tested using a variety of visualization tools for performance analysis, such as the confusion matrix, ROC curve, precision-recall curve, and learning curve. SVM is a margin-based classifier that finds the optimal separating hyperplane in the feature space. Even though SVM is a highly effective classifier for a variety of classification tasks, the application of a linear decision boundary may limit its ability to capture the complex nonlinear relationships that are common in system telemetry data.

As indicated in the results shown in Table II, the SVM model had an accuracy of 0.89282 with precision and recall of 0.69 and 0.83, respectively, resulting in an F1-score of 0.76. These results indicate that even though the model is capable of identifying a large number of cryptojacking events, it tends to have a higher rate of misclassifications compared to ensemble models. The visualization plots shown in Figures 9 and 10 provide additional insights into the classification performance and learning characteristics of the SVM model.

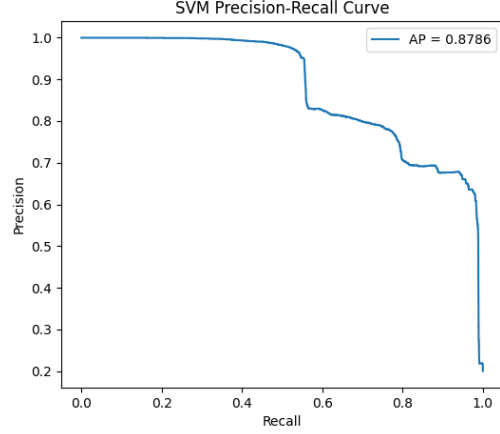


(a) Confusion Matrix

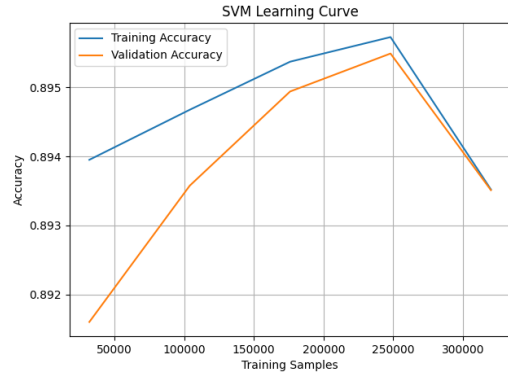


(b) ROC Curve

Fig. 9: Evaluation results of the SVM model (Part 1).



(a) Precision-Recall Curve



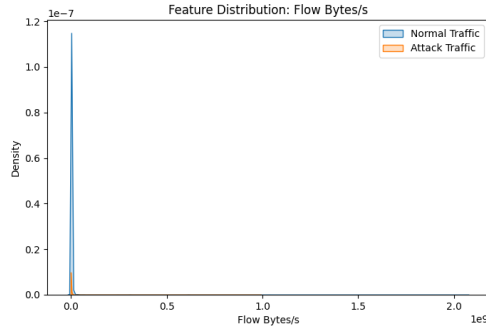
(b) Learning Curve

Fig. 10: Evaluation results of the SVM model (Part 2).

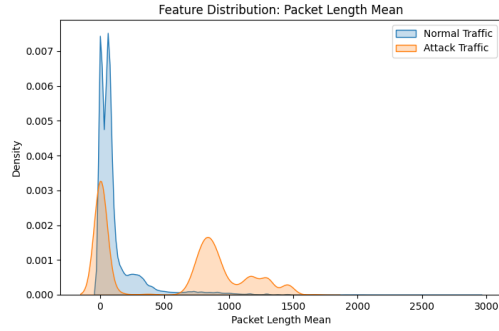
F. Feature Analysis

In order to better understand the role of individual features in the detection of cryptojacking, an in-depth feature analysis was performed. This type of analysis includes the evaluation of feature distributions, the evaluation of the importance of features, and the interpretation of predictions using SHAP values. These three types of analysis provide more in-depth information about the nature of the data and improve the interpretability of the machine learning models. This interpretability is essential for determining the correctness of the detection system and for understanding how individual telemetry feature values influence the detection of cryptojacking.

1) *Feature Distribution*: Analysis of the distribution of feature values was conducted to examine the distribution of different telemetry fields. The knowledge of these distributions is essential in understanding the differences in network activity and discovering possible patterns associated with cryptojacking. The use of feature distributions is also helpful in the identification of skewed data and relationships between key fields in telemetry.



(a) Overall feature distribution



(b) Packet length mean distribution

Fig. 11: Distribution of selected telemetry features used for model training.

2) *Feature Importance*: To identify the attributes that have the most significant influence on the classification of cryptojacking activity, a feature importance analysis was performed. Tree-based ensemble models, such as Random Forest and gradient boosting models, inherently provide a measure of importance for each feature based on the contribution of each feature to the decisions made during the training process. The results show that the features that have the most significant influence on the predictions are those that describe packet length distributions, flow-level features, and bandwidth.

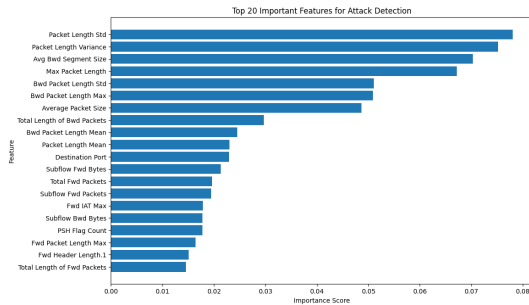
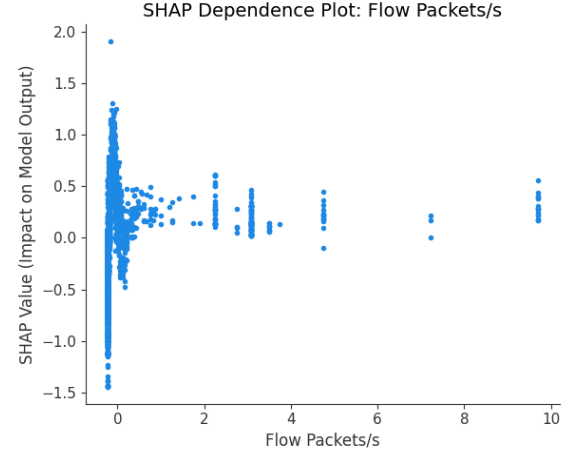
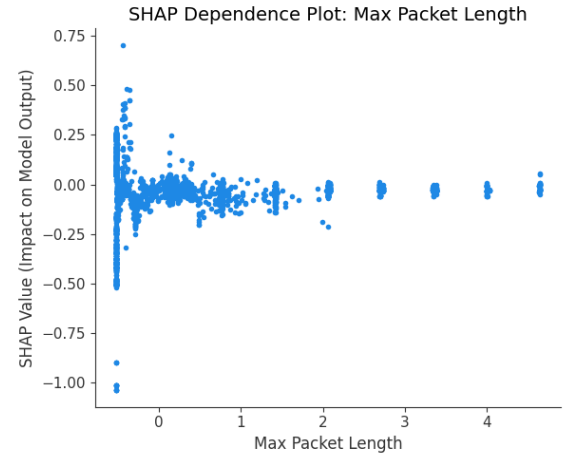


Fig. 12: Feature importance ranking showing the most influential telemetry attributes in the dataset.

3) *SHAP Feature Interpretation*: To enhance the interpretability of the model, a SHAP (SHapley Additive exPlanations) explanation was performed to determine the contribution of each feature to the predictions made by the model. SHAP values provide a unified approach to interpreting machine learning models by quantifying the contribution of each feature to the prediction outcome. From the SHAP explanation, we are able to understand the impact of individual telemetry variables on the detection of cryptojacking events.



(a) SHAP analysis for flow packets



(b) SHAP analysis for maximum packet length

Fig. 13: SHAP-based interpretation of feature contributions in the detection model.

VIII. DISCUSSION

The experimental results indicate that the integration of telemetry signals from multiple layers of the system can greatly improve the effectiveness of cryptojacking detection systems. Traditional detection systems typically rely on a single signal, such as CPU usage or network traffic patterns. Although these approaches can identify mining activities, they are not effective in identifying stealthy cryptojacking attacks

that intentionally consume fewer resources or hide behind encrypted communication channels to remain undetected.

The proposed approach addresses this limitation by integrating indicators of system behavior from multiple sources. By analyzing CPU usage patterns, process behavior, and network traffic patterns simultaneously, the system can identify patterns in multiple dimensions of system behavior. This multi-layered monitoring approach provides a more comprehensive view of system behavior and enables the detection model to identify subtle anomalies indicative of cryptojacking.

Another important strength of this framework is its flexibility. Since the detection logic is designed using machine learning models trained on behavioral characteristics rather than signatures, the framework can adapt to new and evolving attack methods. This is particularly important in the case of cryptojacking attacks, as attackers often modify malware settings to evade conventional security tools.

However, there are a few points that need to be considered before using the framework in a production environment. First, the detection accuracy is highly dependent on the quality and availability of telemetry data. In environments where telemetry data is sparse, noisy, or inconsistent, the accuracy of the model could be impacted. Second, machine learning-based detectors may produce false positives if legitimate high-performance or compute-intensive applications produce patterns similar to cryptocurrency mining.

Some potential ways to enhance the framework would be to incorporate other types of telemetry data, such as disk I/O events or user authentication logs, to improve the detection capabilities. Additionally, integrating the framework with the overall enterprise security monitoring systems could help to enhance the response process.

Conclusion: The discussion highlights the importance of multi-layered behavioral monitoring and machine learning capabilities in enhancing the early detection of cryptojacking attacks.

IX. CONCLUSION AND FUTURE WORK

This paper presented a multi-layer behavioral framework intended for the early detection of cryptojacking malware based on the application of system telemetry and machine learning techniques. The proposed method investigates behavioral clues gathered from multiple layers of the system, such as CPU usage patterns, process behavior, and network communication patterns. By correlating these system telemetry inputs, the system is able to detect unusual system behavior that could potentially indicate the presence of malicious cryptocurrency mining activities on the system.

The experimental results show that the investigation of system behavior based on multiple telemetry inputs enhances the accuracy of cryptojacking malware detection compared to conventional monitoring techniques that focus on a single layer of data. The incorporation of machine learning-based anomaly detection enables the system to detect minute deviations in system behavior while offering a scalable solution for monitoring contemporary computing environments.

Although the proposed framework shows promising detection capability, there are still some areas that offer opportunities for further investigation. For example, the framework could utilize other telemetry data sources like disk IO metrics, authentication logs, and file system access patterns to improve detection capability. In addition, testing the framework in large cloud environments could offer new perspectives on how to detect distributed cryptojacking attacks in virtualized computing environments.

Another possible area of future research is to incorporate automated response capabilities that can isolate malicious processes or limit malicious network communications once malicious activity is identified. This would enable the development of proactive defense systems that can counter cryptojacking attacks before they consume a large amount of computing resources.

In summary, the proposed framework helps to advance intelligent monitoring systems that can detect emerging cryptojacking threats by leveraging behavioral analysis and machine learning approaches.

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